HEAPLINE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

HeapLine product requirements and specifications

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DOCUMENT CONTROL

Document ID	HPL-PRS-001	Version	0.6
Status	Draft	Date	2026-11-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heapline		
Source	docs/03-requirements.md @ 0055524		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-11-03	Amish Chadha	Kit 1.9.1 upgrade: migrated to the PRS format (PR-001 to PR-011 are R1 to R11, same targets; PR-012 to PR-015 added from the user requirements); document ID HPL-REQ-001 becomes HPL-PRS-001; every PR traces to a user requirement in HPL-URS-001
0.5	2026-10-04	Amish Chadha	Siting limit for a shared capstan set decided by Amish (round-3 decision 3A, HPL-DDR-004); targets and results unchanged
0.4	2026-10-03	Amish Chadha	Amish's requirement decisions 20A, 21A, 22A and 23C (HPL-DDR-003): R2 met with the 18 mm tip; R5 met under the four-hauler rule; R8 met in two waves; R9 restated to USD 850 per site search kit with one capstan set shared by neighbouring sites
0.3	2026-10-03	Amish Chadha	TRL 3 results from HPL-CAL-001 on the constructable design (HPL-DDR-002); R2, R5 and R8 at risk and

			R9 not met, with decisions for Amish in docs/REVIEW.md
0.2	2026-10-03	Amish Chadha	TRL 2; concept status for each requirement; R10 (overload limit) and R11 (holds when let go) added under HPL-DDR-001
0.1	2026-09-30	Amish Chadha	Initial scaffold (as HPL-REQ-001)

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What the kit must do, each requirement traced to a user need in the URS (`03a-user-requirements.md`). PR-001 to PR-011 are the earlier requirements R1 to R11 with unchanged targets (column "Was"); PR-012 to PR-015 were added so that every user requirement has a product requirement. Results are paper estimates from HPL-CAL-001 on the constructable design, not test results.

On 2026-10-03 Amish decided the four requirement decisions put to him at TRL 3 as recommended: "i agree with all the 46 recommendations you provided. please proceed." For HeapLine these are 20A (PR-002), 21A (PR-005), 22A (PR-008) and 23C (PR-009), recorded in HPL-DDR-003. On 2026-10-04 Amish decided option 3A on how far apart sites may share a capstan: "For round 3, I agree with all your proposed recommendations". A shared capstan serves only sites within about 500 m walk of the host shed (HPL-DDR-004); farther sites keep their own capstan set or rely on the search kit alone.

> **Safety:** HeapLine is used on unstable ground where a second slide can bury the searchers. It is published as an open engineering reference, not certified rescue equipment. The capstan stores energy in its rope; the overload limit (PR-010) and the holding pawls (PR-011) are safety requirements.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Deploy fast	Kit carried 100 m (330 ft) and first probe line started within 5 minutes of the alarm	5	max	min after the alarm	Timed drills with the partner cooperative	Met on paper	R1
PR-002	UR-002	Performance	Probe depth	Probes reach at least 2.5 m (8.2 ft) into loose waste	2.5	min	m	Probe trials on a waste test heap, including whether the sleeve friction rises with the 18 mm tip's smaller clearance	Met on paper	R2
PR-003	UR-003	Performance	Crawl board support	Board carries a 100 kg (220 lb) searcher with sinkage under 50 mm on loose fill	50	max	mm sinkage	Load tests on representative fill	Met on paper	R3
PR-004	UR-004	Performance	Capstan pull	At least 5 kN (1,124 lbf) pull from a hand capstan on a ground anchor	5	min	kN	Pull test on CalRig	Met on paper	R4
PR-005	UR-005	Performance	Casualty transport	Sheet stretcher moves a 100 kg (220 lb) casualty 30 m (98 ft) over debris with two to four haulers	30	min	m, 100 kg casualty	Drill with manikin	Met on paper under the drill-card rule	R5
PR-006	UR-006, UR-012	Usability	Stop rule understood	All drill participants state the stop signals	100	min	% of participants	Post-drill check	To verify at TRL 4	R6

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				correctly after training						
PR-007	UR-007	Materials	Storage life	All items usable after 12 months stored in the site box	12	min	months	Annual inspection log at the pilot site	Met by design	R7
PR-008	UR-008	Packaging and transport	Portable	No single item over 25 kg (55 lb); whole kit carried by four people	25	max	kg per item	Weighing; timed carry in two waves	Met on paper in two waves	R8
PR-009	UR-009	Performance	Cost	Restated 2026-10-03 (decision 23C): parts at most USD 850 per site search kit, with one capstan set shared by neighbouring sites and costed per group, a reused crate (or bought job box) and PPE from the cooperative's stock. Was: parts under USD 400 per site kit	850	max	USD per site search kit	Costed bill of materials	Met on paper as restated	R9
PR-010	UR-010	Safety	Overload limit	Rope tension limited so the rope, capstan and anchor are never loaded past the 8 kN design tension	8	max	kN rope tension	Shear pin release tests on a batch (TRL 4)	Met on paper	R10
PR-011	UR-010	Safety		The drum cannot run	0	max				R11

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Holds when let go	back when the bars are released			turns of drum run-back	Release test at working pull (TRL 4)	Met on paper	
PR-012	UR-014	Usability	Light without power	Searchers carry head lamps and hand lamps that take AA alkaline cells, with the cells stored out of the lamps	8	min	lamps (six head, two hand)	Inspection at the monthly check (BOM line 25)	Met by design	New
PR-013	UR-011	Materials	Repair with common stock	Every made part is steel tube or bar, timber, board, plastic sheet or rope that a local workshop holds	all	min	made parts	Review of the BOM and HPL-BLD-001	Met by design	PRB constraint
PR-014	UR-013	Safety	Not certified equipment	The README, the drill card and every drawing state that HeapLine is an open engineering reference, not certified rescue equipment	all	min	documents	Document review	Met by design	New
PR-015	UR-015	Performance	Shared capstan arrival	A capstan set kept at a host shed reaches a site within about 500 m walk while the search is under way	15	max	min after the alarm	Timed capstan wave drill at TRL 4	Met on paper	Decision 3A (HPL-DDR-004)

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4. Tolerance "min" means the target is the least acceptable value, "max" the greatest. Requirements met "on paper" are first-order estimates; verification is at TRL 4 or later.

PR-009 as restated (decision 23C): parts at most USD 850 per site search kit, with one capstan set shared by neighbouring sites and costed per group, a reused crate (or bought job box) and PPE from the cooperative's stock. It was: parts under USD 400 per site kit.

Table 2. Results on paper (HPL-CAL-001).

Requirement	Result
PR-001	4.6 min estimated: run to the shed, open, carry the search wave back, post the lookout, lay boards while probes are joined
PR-002	18 mm tip on all six probes (decision 20A). Loosened debris: 2.6 m (full usable depth) by one person. Stiffer debris: 2.6 m by two people, 0.6 m by one
PR-003	44 mm on very loose debris with a 1.5 dynamic factor; bridges a 0.8 m void at 4.0 MPa in the battens
PR-004	5 kN with four people at 114 N each; the shear pin releases at about 6.4 kN (5.1 to 7.6 kN over a batch)
PR-005	Drill-card rule (decision 21A): four haulers on debris, 127 N each on level debris; two haulers only along the boards, 153 N each
PR-006	Six stop signals printed on the lid and on pocket cards
PR-007	Exterior plywood, painted or galvanised steel, UV-stabilised HDPE, acetal, polyester; alkaline cells stored out of the lamps; no lithium cells
PR-008	Heaviest carried item 20.9 kg (capstan frame). Carried in two waves (decision 22A): four people carry the search kit first, 21.5 kg each; a second group of three brings the capstan set, 21.4 kg each
PR-009	USD 827 per site search kit; shared capstan set with its crate USD 547 per group (USD 274 a site if two share it)
PR-010	6 mm S275 shear pin releases at 6.4 kN nominal, 7.6 kN at the top of the batch spread; rope factor 4.5 at 8 kN
PR-011	Two holding pawls on a 24-tooth ring; one pawl takes the design tension at 77 MPa tooth bearing
PR-012	Six head lamps and two hand lamps on AA cells; no lithium cells in the box (BOM line 25)
PR-013	Steel tube and bar, plywood, timber, HDPE sheet, acetal and polyester rope throughout (bom/bom.csv)
PR-014	Safety notes in the PRB, README, drill card and drawing title blocks
PR-015	14.8 min from 500 m walk, 3.8 min from 100 m (HPL-CAL-001 [F4])

Requirements not met: none on paper. PR-006 (stop rule understood) cannot be verified before the drills at TRL 4. The value-engineering target of USD 1,800 (`budget_usd`) is met by the estimated USD 1,374 for the host site.

Assumptions

- Pickers and neighbours will organise a search in the first minutes if they have tools and a plan.
- Burial depths in the first minutes of a slide are within reach of hand probes and shovels in at least some cases.

- Sites or cooperatives can keep a box secure and inspected.
- Formal responders will welcome an organised community search rather than stop it.
- A sound anchor (a tree, a parked vehicle or a structural column) can be found on firm ground near most dumpsites; where none exists the capstan is not used.
- Debris resistance to probing has no published data for dumpsite slides; HPL-CAL-001 brackets it with a loosened and a stiffer case.
- Neighbouring sites that share one capstan set are close enough for its second wave to arrive while the search is still under way; a shared capstan serves only sites within about 500 m walk of the host shed (Amish, 2026-10-04, decision 3A: "For round 3, I agree with all your proposed recommendations"; HPL-DDR-004), checked in the TRL 4 timed drills.
- The cooperative keeps PPE in stock for the kit and replaces what is used.